# HEENT Assessment

NURA 330 Health Assessment | Module 6 Lecture Study Guide

Exam 2 content · reviewed for the final

**Lecture roadmap:** Case: Sarah Mitchell → Why HEENT matters → Anatomy → Cranial nerves → Subjective history → Psychosocial and SDOH → Equipment → Head and face → Eyes (inspection, acuity, visual fields, PERRLA, pupil changes, EOMs) → Ears (inspection, otoscope, hearing, Weber and Rinne) → Nose and sinuses → Mouth and pharynx → Neck, lymph nodes, thyroid, carotids → Red flags → Terminology → Lifespan → Case wrap-up → Documentation

**Exam 2 study guide topics:** HEENT anatomy; **cranial nerves I–XII**; subjective history; SDOH; equipment; head and face; eye inspection and vision; **PERRLA and pupil changes**; **EOMs and the "H" pattern**; ear inspection, otoscopy, hearing; nose and sinuses; mouth and pharynx; neck and trachea; thyroid and carotids; **prioritizing red flags**.

**How to read this guide:** Everything without a tag comes from the lecture slides or the slide notes. Anything else is tagged: STUDY GUIDE emphasized on the exam study guide but not spelled out on the slides; ADDED information not in the slides; CLARIFIED a slide point explained further; CORRECTED an error or inconsistency in the slides. Every tag is also listed in the notes at the end.

**Case: Sarah Mitchell, 58.** "I've been feeling dizzy, and my right ear feels plugged." Intermittent dizziness × 3 days; **room spins when she turns her head**; decreased hearing in the right ear; denies head injury or loss of consciousness; history of seasonal allergies and HTN. VS: BP 138/84, HR 82, RR 16, $SpO_2$ 98%, T 37 °C. Think: what else would you ask, which structures would you assess, and what would need follow-up?

## Why HEENT Matters

HEENT (head, eyes, ears, nose, throat) structures support: **neurological function** (cranial nerves), **vision**, **hearing and balance**, **respiration** (upper airway), **nutrition and swallowing**, and **circulation** (carotids). Changes can point to neurological, respiratory, cardiovascular, or local problems.

## Anatomy

| | |
|---|---|
| **Head & face** | Skull and scalp; facial muscles; **temporomandibular joint (TMJ)**; cranial nerves |
| **Eyes** | Cornea, iris, pupil, lens, retina, optic nerve; extraocular muscles |
| **Ears** | **External:** auricle and canal. **Middle:** tympanic membrane and ossicles. **Inner:** cochlea (hearing) and vestibular system (balance) |
| **Nose & throat** | Nasal cavity and paranasal sinuses; oral cavity, tongue, tonsils, pharynx |
| **Neck** | Lymph nodes, thyroid, trachea, carotid arteries |

## Cranial Nerves

The instructor's note: you do not need every detail. **Focus on linking each assessment maneuver to the nerve it tests.**

| Cranial nerve | Function | How to assess |
|---|---|---|
| **I: Olfactory** | Smell | Identify a familiar scent, one nostril at a time |
| **II: Optic** | Vision | Visual acuity (Snellen) and visual fields (confrontation) |
| **III, IV, VI: Oculomotor, Trochlear, Abducens** | Eye movement (III also constricts the pupil) | EOMs ("H" pattern) and pupil responses |
| **V: Trigeminal** | Facial sensation, chewing | Light touch and sharp/dull on the face; jaw movement against resistance |
| **VII: Facial** | Facial expression, taste | Smile, close eyes tightly, puff cheeks |
| **VIII: Vestibulocochlear** | Hearing, balance | Whisper or finger-rub test; Weber and Rinne |
| **IX, X: Glossopharyngeal, Vagus** | Swallowing, palate movement, voice | Say "ah" (uvula rises midline); voice; swallowing |

| Cranial nerve | Function | How to assess |
|---|---|---|
| **XI: Accessory** | Neck and shoulder movement | Shoulder shrug and head turn against resistance |
| **XII: Hypoglossal** | Tongue movement | Stick out the tongue; it should be midline |

ADDED **Memory aids:** Names in order: "**O**h **O**h **O**h **T**o **T**ouch **A**nd **F**eel **V**ery **G**ood **V**elvet, **A**h **H**eaven." Eye muscles: **LR6 SO4** (lateral rectus = CN VI; superior oblique = CN IV; everything else = CN III).

## Subjective Assessment

Open with: "Have you noticed any recent changes in your vision, hearing, or problems with your head, nose, or throat?" For any positive, ask about onset, duration, severity, associated symptoms, and what makes it better or worse.

| Area | Ask about |
|---|---|
| **Head & face** | Headaches (onset, location, severity, frequency, associated symptoms); dizziness or vertigo; head injury or loss of consciousness |
| **Eyes** | Vision changes, blurred or double vision; pain, redness, discharge; glasses/contacts and last eye exam |
| **Ears** | Hearing loss; ear pain, **tinnitus**, drainage; hearing aids; ear infections |
| **Nose & sinuses** | Congestion, drainage, nosebleeds; sinus pressure; allergies or recurrent sinus infections |
| **Mouth & throat** | Sore throat, hoarseness, voice changes; trouble swallowing or choking; dental pain, mouth sores, dry mouth |
| **Relevant history** | Prior HEENT conditions or surgery; meds (eye/ear drops, allergy meds); family history of hearing or vision problems; tobacco/vaping, **occupational noise**, oral hygiene |

### Psychosocial and SDOH

- Vision or hearing loss affects communication and independence; chronic dizziness can cause **fear of falling**; changes in appearance, speech, or swallowing can affect confidence; chronic symptoms bring frustration or anxiety.

- **SDOH:** access to vision and hearing screening; cost of glasses, hearing aids, dental care; noise or irritant exposure at work; transportation to specialists; health literacy and language barriers.
- **Nursing:** ask about barriers; provide communication accommodations; connect to follow-up resources. Untreated hearing loss in the hospital is a **safety risk** (missed instructions, alarms, consent).

# Objective Assessment

**Equipment:** penlight, otoscope, tongue depressor, Snellen chart, tuning fork, stethoscope, gloves/PPE. **Techniques:** inspection, palpation, auscultation, functional testing. **Order:** head → eyes → ears → nose → mouth/throat → neck. **Compare both sides throughout.**

## Head and Face

- **Inspect:** skull shape, size, symmetry; hair distribution, texture, hair loss; lesions, swelling, masses; facial symmetry and involuntary movements.
- **Palpate:** skull for tenderness or masses; **TMJ** while the patient opens and closes the mouth (tenderness, clicking, limited movement).
- **CN VII:** smile, close eyes tightly against gentle resistance, puff cheeks; movement should be equal on both sides.
- **CN V:** light touch and sharp/dull on both sides of the face with the patient's eyes closed; jaw side to side against resistance.

# Eye Assessment

## Inspection and Vision (CN II)

- **External:** lids and lashes; conjunctiva and sclera; pupil size, shape, equality; redness, swelling, drainage. **Expected:** white sclera, conjunctiva without abnormal redness.
- **Visual acuity:** Snellen chart at **20 feet**, **one eye at a time**; document with or without glasses.
- **Visual fields (confrontation):** face the patient about 2 feet away at eye level. Patient covers one eye, you cover your opposite eye. Patient looks at your open eye. Bring your fingers in from the edges in all four quadrants; the patient says when they see them. Repeat for the other eye. Their fields should match yours.

ADDED **Reading a Snellen result:** 20/40 means the patient reads at 20 feet what a person with normal vision reads at 40 feet. **The larger the bottom number, the worse the vision.** 20/20 is normal.

## PERRLA

### Pupils Equal, Round, Reactive to Light and Accommodation

| Step | How | Expected |
|---|---|---|
| **Size & shape** | Compare pupils; measure in mm | Equal and round |
| **Direct light response** | Shine a penlight into one eye | **That** pupil constricts |
| **Consensual light response** | Shine light into one eye; watch the **other** | The **opposite** pupil also constricts |
| **Accommodation** | Patient looks at a distant object, then at your finger as you move it toward their nose | Pupils **constrict** and eyes **converge** (cross) |

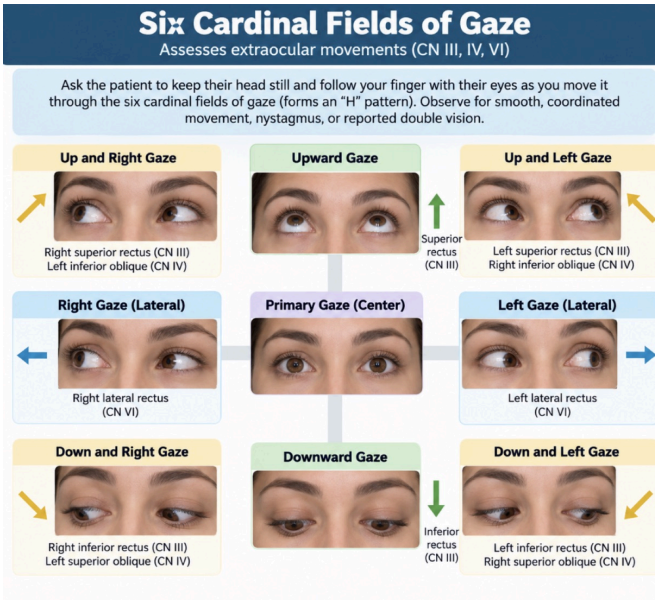
The light reflex tests **CN II** (senses the light) and **CN III** (constricts the pupil). New unequal or nonreactive pupils may mean neurological dysfunction.

## Pupil Changes

| Finding | Meaning |
|---|---|
| **Brisk** | **Normal** |
| **Sluggish** | Compression of the oculomotor nerve (CN III): **early herniation**, cerebral edema |
| **Fixed / nonreactive (midposition)** | Loss of both sympathetic and parasympathetic input: midbrain infarction or herniation |
| **Small, pinpoint, nonreactive** | Damage to the **pons** (hemorrhage, ischemia) or **opioid overdose** |

## Extraocular Movements (CN III, IV, VI)

- Patient keeps the **head still**; hold your finger **12–18 inches** away; move it through the **six cardinal fields of gaze (the "H" pattern)**.
- Watch both eyes for smooth, coordinated movement; ability to follow; **nystagmus** (involuntary rhythmic movement); **diplopia** (double vision).
- **Expected:** eyes move together smoothly; no abnormal nystagmus or double vision.




*Six cardinal fields of gaze (lecture slide)*

# Ear Assessment

- **Inspect** the auricles (helix, tragus, lobule) for symmetry, position, lesions, drainage. Use the **otoscope** for the canal and tympanic membrane.
- **Palpate** the auricle and tragus for tenderness or swelling.
- **Hearing (CN VIII):** whisper test or finger rub, **one ear at a time**.
- **Expected:** symmetric ears; no tenderness or drainage; clear canals; **tympanic membrane intact and pearly gray**; hearing intact bilaterally.

ADDED **Otoscope technique:** Use the largest speculum that fits. For adults, pull the auricle **up and back** to straighten the canal; for children under about 3, pull **down and back**. Hold the otoscope so your hand rests against the patient's head, in case they move.

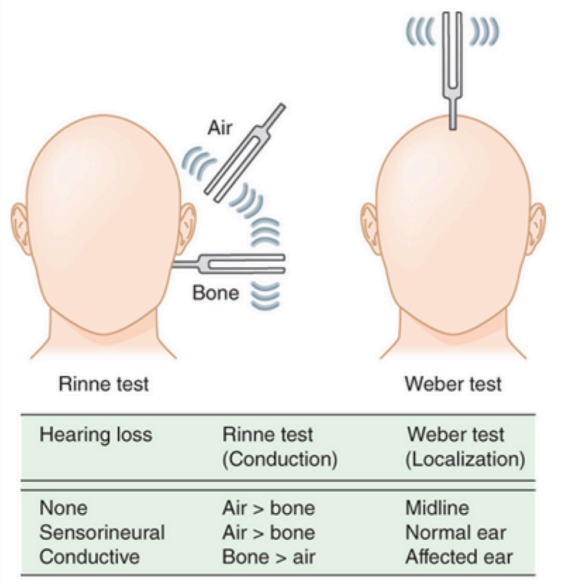
## Weber and Rinne Tests

Help tell **conductive** from **sensorineural** hearing loss.

| Test | How | Normal |
|---|---|---|
| **Weber** | Vibrating tuning fork on the **midline of the forehead/head**; "Where do you hear it?" | Heard **equally in both ears** (midline) |

| Test | How | Normal |
|---|---|---|
| **Rinne** | Fork on the **mastoid bone**; when sound stops, move it next to the ear canal | **Air conduction > bone conduction (AC > BC)** |

| Type of loss | Problem | Weber | Rinne |
|---|---|---|---|
| **Conductive** | Sound cannot get through the outer/middle ear: **earwax**, ear infection, fluid, perforated eardrum, ossicle damage. Sounds muffled. Often treatable | Louder in the **affected** ear | **BC ≥ AC** |
| **Sensorineural** | Damage to the inner ear (cochlea) or CN VIII: **aging (presbycusis)**, loud noise, ototoxic meds. Speech hard to understand, especially in noise. Usually permanent | Louder in the **better (normal)** ear | AC > BC (still) |




| Hearing loss | Rinne test (Conduction) | Weber test (Localization) |
|---|---|---|
| None | Air > bone | Midline |
| Sensorineural | Air > bone | Normal ear |
| Conductive | Bone > air | Affected ear |

*Rinne and Weber tests (lecture slide)*

## Nose, Sinuses, Mouth, and Pharynx

| Area | Assess | Expected |
|---|---|---|
| **Nose** | External symmetry or deformity; **patency one nostril at a time**; mucosa and septum with light/otoscope; drainage, bleeding, inflammation, septal deviation | Symmetric; nares patent; **mucosa pink and moist**; no drainage or bleeding |
| **Sinuses** | Palpate **frontal** and **maxillary** sinuses for tenderness | Nontender (tenderness suggests inflammation, infection, allergies) |
| **Smell (CN I)** | If the patient reports trouble smelling: one nostril at a time, familiar non-irritating scent | Identifies scent |
| **Lips & mucosa** | Color, moisture, lesions; gums and teeth | Pink, moist, no lesions |
| **Tongue (CN XII)** | Top, sides, underside; movement and symmetry; taste changes if reported | **Midline**, symmetric movement |

| Area | Assess | Expected |
|---|---|---|
| **Pharynx & uvula (CN IX, X)** | Tonsils and back of throat (redness, swelling, exudate); say "ah" | **Uvula midline; soft palate rises equally** |
| **Swallowing & voice** | Hoarseness, speech changes, trouble swallowing or choking | No difficulty |

# Neck Assessment

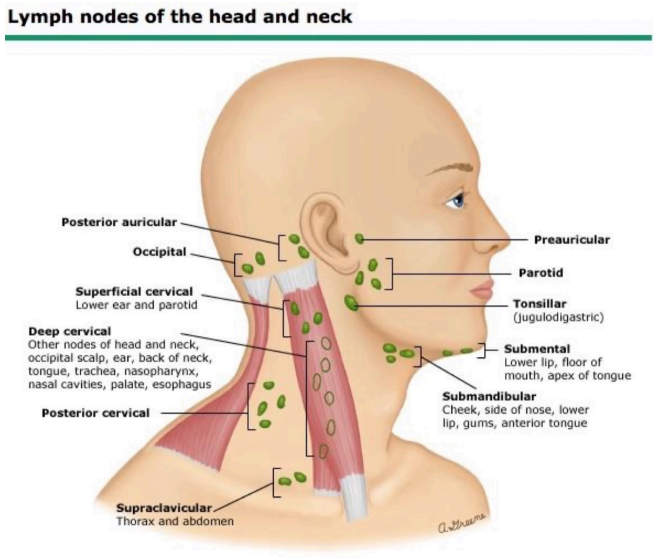
| Inspect | Palpate |
|---|---|
| Symmetry and alignment; masses, swelling, scars, abnormal movements; **range of motion**; **tracheal position**; **JVD** | **Trachea midline**; cervical lymph nodes systematically; **thyroid while the patient swallows**; **carotid pulses one at a time**. CN XI: shrug and head turn against resistance |

**Expected:** symmetric with full ROM; **trachea midline**; no masses; no JVD when positioned correctly; no thyroid enlargement.

## Lymph Nodes

Palpate **both sides** with gentle, circular motions using the finger pads. Note **size, tenderness, consistency, mobility, location**.

- **Expected:** not palpable, or small, soft, mobile, nontender.
- **Abnormal:** enlarged, firm, fixed, or tender nodes need further evaluation (see Module 4: tender + mobile suggests infection; hard + fixed + painless suggests malignancy).




*Lymph nodes of the head and neck (lecture slide)*

## Thyroid and Carotids

- **Thyroid:** inspect the front of the neck; palpate from **behind (posterior approach)**; have the patient **swallow** (the thyroid moves up); note enlargement, nodules, tenderness. **If enlarged, auscultate for a bruit.**
- **Carotids:** palpate gently, **one side at a time**; check strength and symmetry; auscultate for bruits when indicated.
- **Bruit** (broo-EE): a swishing or whooshing sound over an artery from **turbulent blood flow**, usually from narrowing (stenosis, atherosclerosis). Normal arteries make no sound. Also listened for over the abdominal aorta and renal arteries.
- **Expected:** thyroid not enlarged, no nodules; carotids palpable and equal (2+); no bruits.

ADDED **Technique tip:** Listen for a carotid bruit with the **bell** (low-pitched sound) while the patient briefly holds their breath, so breath sounds do not interfere.

## Prioritizing Unexpected Findings

| Urgent: assess now and escalate | Needs further evaluation |
|---|---|
| **Sudden facial droop or asymmetry**<br>**New unequal pupils with neuro symptoms**<br>**Sudden vision loss**<br>**Sudden severe headache**, especially with neuro changes<br>**Trouble swallowing with drooling, stridor, or respiratory distress**<br>**New tracheal deviation with respiratory compromise**<br>**Sudden hearing loss** | Persistent hoarseness<br>Enlarged or fixed lymph nodes<br>Persistent mouth sores<br>Recurrent nosebleeds<br>Ear pain or drainage<br>Thyroid enlargement or nodules |

**Classic question:** A patient develops **sudden facial drooping and slurred speech** during your assessment. First action? **Activate the emergency/stroke response immediately** while assessing the airway and other neuro findings.

## Medical Terminology

| Term | Meaning |
|---|---|
| **PERRLA / EOM** | Pupils equal, round, reactive to light and accommodation / extraocular movements |
| **Anisocoria** | Unequal pupil sizes |
| **Nystagmus** | Involuntary rhythmic eye movement |
| **Diplopia** | Double vision |
| **Ptosis** | Drooping upper eyelid |

| Term | Meaning |
|---|---|
| **Tinnitus** | Ringing or other noise in the ears |
| **Vertigo** | Sensation of spinning or movement |
| **Epistaxis** | Nosebleed |
| **Dysphagia / Dysphonia** | Difficulty swallowing / abnormal voice |
| **Lymphadenopathy** | Abnormal lymph node enlargement or characteristics |
| **Thyromegaly** | Enlarged thyroid |
| **Bruit** | Abnormal vascular sound from turbulent flow |
| **JVD** | Jugular venous distention |

## Lifespan Considerations

| Infants & children | Adults | Older adults |
|---|---|---|
| Assess **fontanels** and skull development; age-appropriate vision and hearing tests; **ear infections more common**; tonsils and nodes may be more prominent; speech, hearing, and vision milestones | Occupational and recreational exposures; screen for vision and hearing changes; tobacco and chronic conditions | **Presbyopia** (trouble focusing up close); **presbycusis** (age-related hearing loss); ↑ cataracts; dentition and oral health changes; **reduced taste and smell** affect nutrition and safety |

## Back to Sarah Mitchell

**Findings:** A&O; PERRLA; EOMs intact; no facial droop or speech change; **right canal partly blocked by cerumen**; right TM not fully seen; **decreased hearing on the right (whisper test)**; brief spinning with position changes; gait steady.

| Unexpected findings | Additional assessment | Next steps |
|---|---|---|
| Decreased hearing, ear fullness, cerumen obstruction, positional vertigo | Further describe the dizziness; orthostatic VS if indicated; focused neuro assessment; more hearing testing | **Fall precautions**; report findings to the provider; **do not assume the earwax explains the dizziness** |

## Documentation

**Normal HEENT (from the slides):** "Head normocephalic, atraumatic. Facial features symmetrical; facial sensation intact bilaterally. PERRLA, EOM intact. Sclera white, conjunctiva without redness. Hearing grossly intact bilaterally. External ears without lesions or drainage. Nares patent bilaterally; nasal mucosa pink and moist. Oral mucosa pink and moist; tongue and uvula midline. Neck supple with full ROM; trachea midline. No palpable cervical lymphadenopathy or thyroid enlargement. Carotid pulses 2+ bilaterally."

**Abnormal example:** "Patient reports decreased hearing and fullness in right ear for three days. Right external auditory canal partially obstructed by cerumen; tympanic membrane incompletely visualized. Decreased hearing noted in right ear during whisper test. Patient reports intermittent spinning sensation with position changes. Provider notified."

Documentation should be objective and specific, accurate and concise, based only on what you actually assessed, and separate subjective from objective findings.

## Quick Review

| Topic | Key points |
|---|---|
| **Order** | Head → eyes → ears → nose → mouth/throat → neck; compare sides |
| **CN tests** | I smell; II acuity/fields; III/IV/VI EOMs and pupils; V face sensation/jaw; VII smile, puff cheeks; VIII hearing; IX/X "ah"; XI shrug; XII tongue |
| **Snellen** | 20 feet, one eye at a time, note glasses |
| **PERRLA** | Direct + consensual constriction; accommodation = constrict + converge |
| **Pupils** | Sluggish = early herniation; fixed midposition = midbrain/herniation; pinpoint = pons or opioids |

| Topic | Key points |
|---|---|
| **EOMs** | Six cardinal fields ("H"), 12–18 in., head still; watch for nystagmus, diplopia |
| **Ear** | TM pearly gray, intact; whisper test one ear at a time |
| **Weber / Rinne** | Normal: midline / AC > BC. Conductive: affected ear / BC ≥ AC. Sensorineural: good ear / AC > BC |
| **Neck** | Trachea midline; thyroid from behind while swallowing; carotids one at a time; bruit = turbulent flow |
| **Red flags** | Sudden facial droop, unequal pupils with neuro signs, vision or hearing loss, worst headache, drooling/stridor, tracheal deviation |

## Self-Check ADDED

1. The patient's uvula deviates to one side when saying "ah." Which nerves? *Answer: CN IX and X.*
2. Light is shone in the right eye and the left pupil constricts. What is this called? *Answer: The consensual response.*
3. Weber lateralizes to the left ear, and Rinne on the left shows BC > AC. Type of hearing loss? *Answer: Conductive loss in the left ear.*
4. A patient has pinpoint, nonreactive pupils and slow respirations. What should the nurse suspect? *Answer: Opioid overdose (or pontine injury).*
5. How is the thyroid palpated? *Answer: From behind, while the patient swallows.*
6. Why should the nurse not palpate both carotids at once? *Answer: It can reduce blood flow to the brain.*

## Notes on Sources, Additions & Clarifications

**Corrections:** none needed; the HEENT slides were internally consistent.

**From the Exam 2 study guide:** the topic list in the blueprint box.

**Added information (not in the slides):**

- Cranial nerve names mnemonic and LR6 SO4.
- How to read a Snellen fraction.
- Otoscope technique (pull up and back for adults, down and back for young children).
- Using the bell and a breath-hold for carotid bruits.
- The point that untreated hearing loss is a hospital safety risk is the answer to the slide's own discussion question, not a slide statement.
- The self-check questions.

**Other:** The Weber/Rinne interpretation table combines the slide notes and the slide's image. The slide notes say the consensual reflex tests CN II and CN III; this guide applies that to the light reflex generally.